

What went wrong while calculating the state $\psi = 1/\sqrt{2}(|011\rangle - |100\rangle)$

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I've been trying to create the state $\psi = 1/\sqrt{2}(|011\rangle - |100\rangle)$, but in qiskit it is generating $\psi = 1/\sqrt{2}(|001\rangle - |110\rangle)$ instead. Is this type of things happen in qiskit, or my calculation is wrong, or is it because of the noise. I'm unable to detect what went wrong.

I've 1st initialize the 3 qubit system with $|0\rangle$ state, then have applied H gate on qubit-0, then have applied X gate on qubit-1, then CNOT gate on qubit-0 (control), and on qubit-1 (target), then again CNOT gate on qubit-1 (control), and on qubit-2 (target), and at last Z gate to make it negative. My [calculation](#) shows that it was able to successfully create the $\psi = 1/\sqrt{2}(|011\rangle - |100\rangle)$ state,

the Calculations that I've done:

To make the state $\psi = (|011\rangle - |100\rangle)/\sqrt{2}$:

First applying H gate to qubit-0 in a 3 qubit system, so the output becomes: $K(|000\rangle) \Rightarrow (|000\rangle + |100\rangle)/\sqrt{2}$

Then applying X-gate to qubit-1, so the output becomes: $X((|000\rangle + |100\rangle)/\sqrt{2}) \Rightarrow (|010\rangle + |110\rangle)/\sqrt{2}$

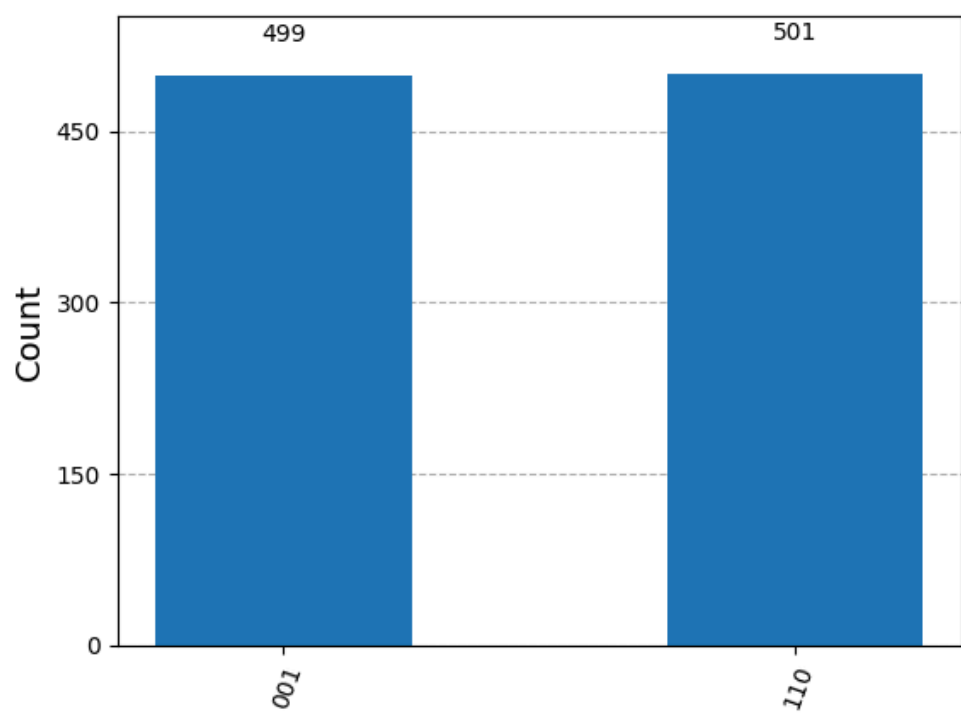
Then applying CNOT gate to qubit-0 as control, and qubit-1 as target, so: $CNOT((|010\rangle + |110\rangle)/\sqrt{2}) \Rightarrow (|010\rangle + |100\rangle)/\sqrt{2}$

Then again applying CNOT gate to qubit-1 as control, and qubit-2 as target, so: $CNOT((|010\rangle + |100\rangle)/\sqrt{2}) \Rightarrow (|011\rangle + |100\rangle)/\sqrt{2}$

Then applying Z gate on the qubit-0, so it becomes: $Z((|011\rangle + |100\rangle)/\sqrt{2}) \Rightarrow (|011\rangle - |100\rangle)/\sqrt{2}$ which is the expected state

```
qc = QuantumCircuit(3) #creating a 3-qubit system
qc.h(0) #applying H gate to make entagled state
qc.x(1) #X gate: if 0, then 1; if 1, then 0
qc.cx(0,1) #if control 0, then target no change; if control 1, then will flip the target
qc.cx(1,2)
qc.z(0) #if 0, then no change; if 1 then will make it -1
```

but in qiskit, I've created the circuit that way, have measured it and plotted histogram and have run it on QASM simulator, but it was showing the wrong entangled state, which is the following:



quantum-computing qiskit Edit tags

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asked Nov 24 at 15:37
SARA_1729
3 5

1 Answer

Sorted by: Date modified (newest first)

Qiskit orders qubits from right to left, that is, qubit 0 is the rightmost and so on, so the last qubit (qubit 2 in your case) is the leftmost. You can use `Statevector` to check the state you have created

```
from qiskit.quantum_info import Statevector
sv = Statevector.from_instruction(qc)
display(sv.draw('latex', prefix='\\psi \\rangle ='))
```

If you prefer to see the final state with qubits in the usual order, you can use `qc.reverse_bits()` which reverses qubits order.

answered Nov 26 at 13:33
luc25
16 1